

Plain language summary

Tourette syndrome is a condition that causes repeated movements and sounds called tics. Medicines currently used to treat tics can be effective, but they may also cause unwanted side effects such as weight gain, sleepiness, and movement problems. Ecopipam is a new medicine that works differently from currently available treatments. We reviewed and combined the results of three clinical trials involving 297 people with Tourette syndrome to better understand how well ecopipam works and how safe it is. Our analysis showed that people taking ecopipam experienced greater improvements in tic severity than those taking placebo (an inactive treatment). The benefits were consistent across studies. We also found that common side effects such as headache and insomnia were not clearly more frequent with ecopipam than with placebo, although the available safety data were limited. Overall, the current evidence suggests that ecopipam may be an effective and generally well-tolerated treatment for Tourette syndrome. However, additional large studies with longer follow-up are needed to confirm its long-term benefits and safety.

Keywords

ecopipam, Tourette syndrome, tic disorder, dopamine D1 receptor antagonist, YGTSS, meta-analysis, randomized controlled trial

Received: 2 June 2026; Revised: 17 July 2026; Accepted: 27 August 2026

Introduction

Tourette syndrome is defined by multiple motor tics and at least one vocal tic persisting for more than 1 year, with onset before adulthood.¹ Tic expression varies across development and may fluctuate with stress, arousal, fatigue, and contextual demands. Although many patients improve during adolescence, a clinically important subgroup has persistent tics associated with pain, stigma, social avoidance, sleep disruption, academic impairment, and reduced quality of life.^{2,3} A population-based systematic review has also characterized the prevalence of Tourette syndrome across studied populations.⁴ The Yale Global Tic Severity Scale (YGTSS) remains the principal clinician-rated endpoint in therapeutic trials because it captures motor and vocal tic number, frequency, intensity, complexity, and interference.⁵

Current treatment is individualized. Behavioural therapy, particularly comprehensive behavioural intervention for tics, is recommended when available and acceptable.^{6,7} Pharmacological options include alpha-2 adrenergic agonists, antipsychotics, and vesicular monoamine transporter 2 inhibitors, but no medication is ideal for all patients.^{8,9} Dopamine D2 receptor-blocking agents can reduce tics, yet their use is often limited by somnolence, akathisia, parkinsonism, weight gain, metabolic effects, prolactin elevation, and concerns about tardive syndromes during longer treatment.^{10,11} These limitations support continued evaluation of targeted dopaminergic strategies that may preserve anti-tic efficacy while improving tolerability.

Ecopipam is an orally administered selective dopamine D1/D5 receptor antagonist. The therapeutic rationale is based on the role of cortico-striato-thalamo-cortical circuits and dopaminergic signalling in tic generation.^{12,13} An early open-label adult study suggested tic reduction and acceptable short-term tolerability, providing a basis for randomized paediatric evaluation.¹⁴ Subsequent randomized trials included a placebo-controlled crossover trial, a larger 12-week parallel-group trial, and a phase 3 randomized-withdrawal study evaluating maintenance of effect.^{15–17} Because randomized evidence has recently expanded, a focused synthesis is needed for clinicians and investigators. We therefore performed an updated systematic review and meta-analysis of randomized controlled trials of ecopipam for Tourette syndrome, emphasizing efficacy, headache, insomnia, and robustness of the primary outcome.

Two previous systematic reviews evaluated ecopipam for Tourette syndrome. Panda et al. reported a significant reduction in YGTSS-TTS based predominantly on the two earlier randomized trials, whereas Darwish et al. reported a nonsignificant week-4 difference in YGTSS-TTS but improvements in selected global clinical measures. Both reviews were based on searches conducted before publication of the 2026 phase 3 randomized-withdrawal trial. An updated synthesis restricted to randomized placebo-controlled evidence and incorporating the phase 3 data was therefore warranted.^{18,19}

Methods

Protocol and reporting standards

This systematic review and meta-analysis was conducted according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 statement and recommendations from the Cochrane Handbook for Systematic Reviews